

Complexity-Based Market Regime Detection for Dynamic Asset Allocation

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Abstract

This project develops a complexity-based market regime detection framework for dynamic asset allocation between NIFTYBEES and GILT5YBEES. Market behaviour is characterized using four nonlinear complexity measures: Hurst Exponent, Largest Lyapunov Exponent, Sample Entropy, and Lag-1 Autocorrelation. These metrics are used to classify the market into three regimes—Equilibrium, Complete Chaos, and Edge-of-Chaos (Momentum)—which drive portfolio allocation decisions. A volatility-targeting overlay, regime persistence filter, and realistic transaction cost model are incorporated to improve practical applicability. The framework is calibrated using historical data from 2014–2022 and evaluated through an out-of-sample walk-forward test from 2023–2026. Results indicate that the regime-aware strategy outperformed a passive NIFTYBEES benchmark, achieving higher risk-adjusted returns and lower drawdowns. The findings suggest that complexity metrics can provide useful information about changing market conditions and support more resilient portfolio allocation decisions.

1. Introduction

Financial markets frequently break trends and sustained behavior do not last long. Episodes of heightened uncertainty, volatility, and structural change are commonly seen in financial time series. Traditional portfolio construction methods do not account for changing regimes and are thereby vulnerable to sudden shocks, larger drawdowns and higher volatility.

This project uses complexity-based metrics to detect regimes and accordingly allocate capital between equity and bonds that maximizes returns during trending periods and minimizes risks during structural changes.

This methodology uses complexity metrics to understand the market dynamics and classify regimes, which is then used for portfolio allocation.

Instead of classifying market regimes into bull, bear and no trend, this project classifies regimes into: equilibrium (mean reverting), complete chaos (high volatility) and, edge-of-chaos (momentum). This classification instantly makes it useful for portfolio allocation decisions.

The framework is implemented using two Indian exchange-traded funds:

- **NIFTYBEES** – Equity market exposure tracking the Nifty 50 Index.
- **GILT5YBEES** – Government bond exposure serving as the defensive asset.

The primary objective is to improve risk-adjusted performance and reduce drawdowns through regime-aware asset allocation.

2. Methodology

Complexity Metrics

Market behavior is represented using four complementary complexity measures computed on rolling windows of daily log returns of NIFTYBEES only:

1. **Hurst Exponent (H)**

Measures long-term persistence and trend structure. Generally:

- $H > 0.5$ indicates persistent behavior.
- $H \approx 0.5$ indicates random-walk dynamics.
- $H < 0.5$ indicates mean-reverting behavior.

2. **Largest Lyapunov Exponent (LLE)**

Measures the divergence of nearby trajectories in reconstructed phase space. Higher values indicate greater instability and a higher sensitivity to small perturbations.

3. **Sample Entropy (SampEn)**

Quantifies the irregularity and unpredictability of returns. Larger values correspond to more influx of information into the system and therefore greater complexity and disorder.

4. **Lag-1 Autocorrelation (Rho)**

Captures short-term persistence or mean-reversion in returns by analyzing linear relationships.

These metrics jointly form a multidimensional market state representation:

$Z_t = [\text{Hurst Exponent}_t,$
 Largest Lyapunov Exponent_t,
 Sample Entropy_t,
 Lag-1 Autocorrelation_t]

which characterizes the prevailing market environment.

3. Regime Classification Framework

The project classifies market behavior into three economically interpretable regimes:

State 0 – Equilibrium Regime

- Low complexity
- Mild mean reversion
- Relatively stable market structure

Allocation Implication:

- Defensive positioning
- Moderate equity exposure

State 1 – Complete Chaos Regime

- High entropy
- Elevated instability
- Weak predictability

Allocation Implication:

- Exit or significantly reduce equity exposure
- Shift capital toward government bonds

State 2 – Edge-of-Chaos / Momentum Regime

- Persistent market behavior
- Structured information flow
- Strong trend continuation

Allocation Implication:

- Aggressive equity allocation
- Maximum participation in market trends

To prevent excessive switching between states, a persistence filter is applied, requiring regime changes to remain consistent for a minimum duration before implementation.

In-sample rolling window was applied to all four metrics and their values were calibrated to give a satisfactory model fit.

In-sample classification of regime is given in Fig. 1 below.

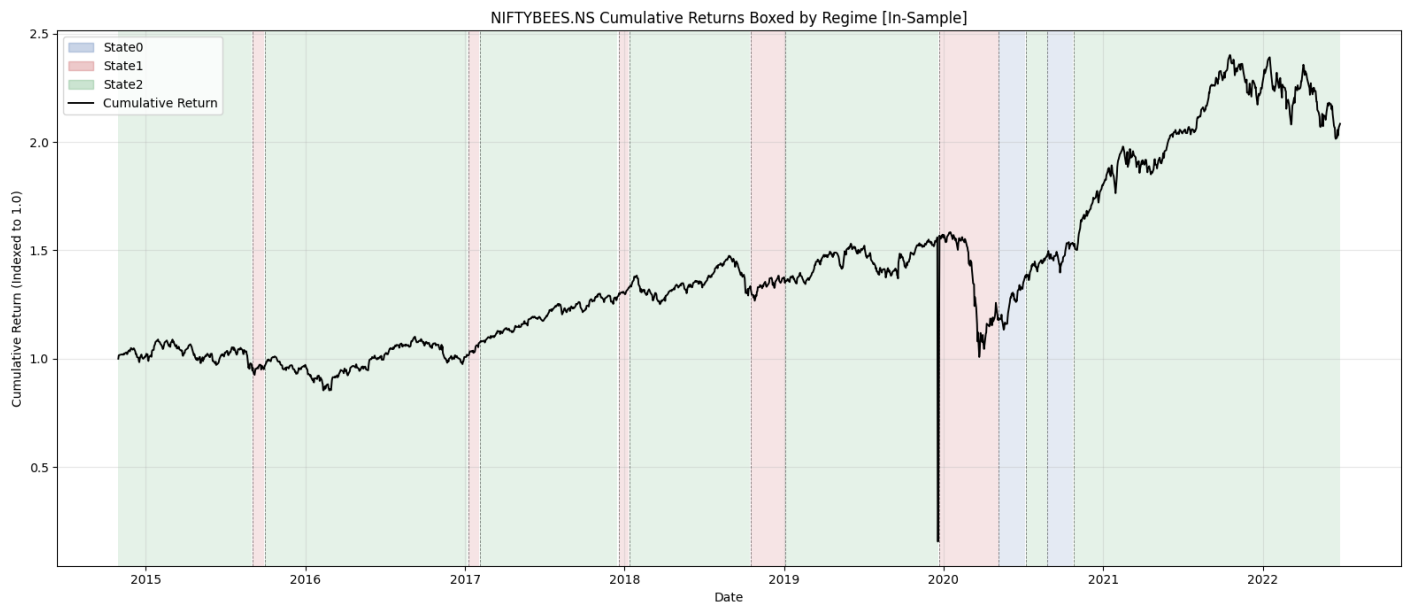


Fig. 1: In-Sample results from Nifty BEES (Jan. 2014 till Jun. 2022) is used to build, test and calibrate the model parameters. In order to avoid overfitting, some degree of error in regime detection was tolerated.

In-sample evaluation of the model during some important market events, where values are represented as percentage of days classified within a regime:

Event	% days in State 0	% days in State 1	% days in State 2
Covid (Feb. – Apr. 2020)	0.00%	100.00%	0.00%
Bull market of 2017 (Jan. 2017 – Dec. 2017)	0.00%	12.20%	87.80%
IL&FS (Sept. 2019 – Dec. 2019)	0.00%	69.20%	30.80%
Calm 2019 (Jan. 2019 – Jun. 2019)	0.00%	0.00%	100.00%

Note: IL&FS event (2019) was primarily a credit and liquidity event. Its impact in equity market was a sustained directional drawdown without a sudden spike. It produced high Hurst Exponent (high persistence of a directional movement), moderate Lyapunov Exponent (trajectories were not diverging explosively), moderate Sample Entropy (lack of sudden information influx) and a positive lag-1 autocorrelation (each drawdown followed by more drawdown). By definition this matches the requirements for a State 2 regime, just that it is a downward momentum.

4. Portfolio Construction

The detected regime determines the allocation between NIFTYBEES and GILT5YBEES.

The portfolio incorporates:

- Regime-dependent asset allocation
- Volatility targeting
- Position constraints
- Realistic transaction costs

The model is calibrated using historical data from January 2014 to June 2022.

An embargo period from July 2022 to December 2022 is excluded from both training and testing to reduce information leakage.

The final evaluation is conducted out-of-sample (walk-forward test) from Jan. 2023 till May 2026.

Regime Allocation Bounds

The allocation framework separates two decisions: (1) which strategic tier the current regime warrants, and (2) how much within that tier given current volatility.

State	Regime	Equity Weight Bounds	Bond Weight
State 0	Equilibrium	1% – 15%	85% – 99%
State 1	Complete Chaos	1% – 40%	60% – 99%
State 2	Momentum	75% – 100%	0% – 25%

Within each regime, a volatility-targeting rule scales the equity weight to a daily volatility target of 0.012 or an approx. 19% annual volatility target.

The bond weight is always calculated as: $\text{weight}_{\text{bond}} = 1 - \text{weight}_{\text{equity}}$.

This is a fully invested and unlevered portfolio.

Transaction Cost Model

NSE delivery-based transactions incur the following costs (2026 rates):

Cost Component	Equity Buy	Equity Sell	Bond Buy	Bond Sell
STT	0.10%	0.10%	0%	0%
Stamp Duty	0.015%	0.00%	0.015%	0.00%
Exchange + GST	~0.0035%	~0.0035%	~0.0035%	~0.0035%
Total	~11.85 bps	~10.35 bps	~1.85 bps	~0.35 bps

No STT is levied on debt-oriented ETFs such as GILT5YBEES.NS. Brokerage is zero on delivery trades for discount brokers. Capital gains tax is excluded, consistent with standard practice, since CGT liability is investor-specific (tax slab, holding period, prior-year losses).

At every rebalance, costs are applied symmetrically: buying equity requires selling bonds (both legs charged), and vice versa.

5. Results

The regime-aware allocation strategy was combined with transaction cost drags and was compared against a passive buy-and-hold strategy for Nifty BEES.

Out-of-Sample Performance (2023–2026)

Metric	Regime Strategy	Buy & Hold (Nifty BEES only)
Annualized Return	13.53%	9.62%
Annualized Sharpe Ratio	1.33	0.83
Maximum Drawdown	-10.51%	-15.23%
Calmar Ratio	1.29	0.63

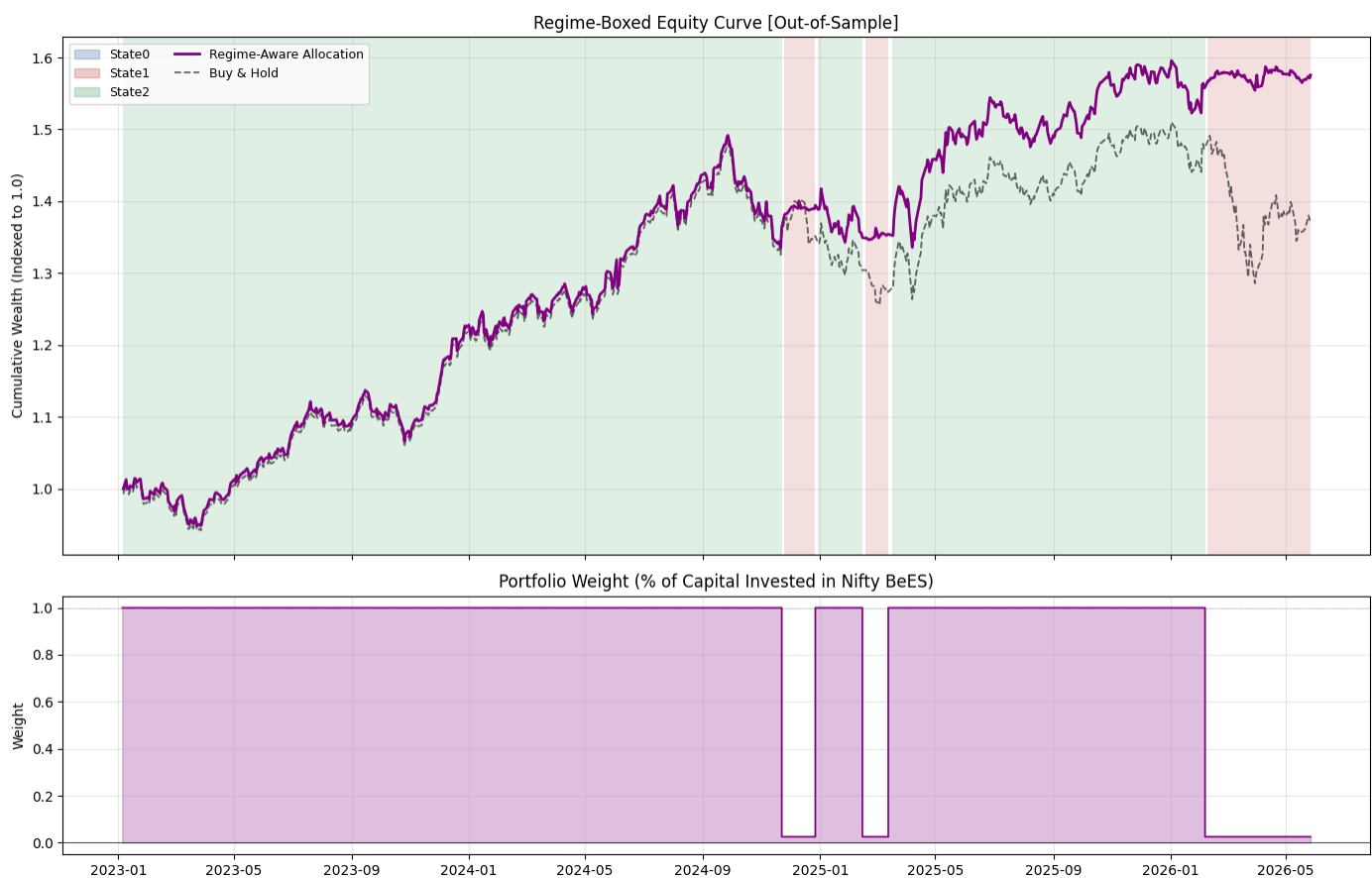


Fig. 2: Top graph: Walk-forward test results between Jan. 2023 till May 2026 for the portfolio. Bottom graph: Weight allocated to Nifty BEES.

Note: Weight allocated to GILT5YBEES = 1 – Weight allocated to Nifty BEES.

Key Findings

- The strategy achieved a significant improvement in annualized returns by approx. +3.90% after subtracting transaction costs.
- The strategy significantly reduced Nifty BEES exposure during State 1 regimes, which usually have a higher volatility.
- Risk-adjusted performance improved significantly, as reflected by the higher Sharpe Ratio of 1.33 which is +0.5 buy-and-hold strategy.
- Maximum drawdown was reduced by approximately -4.72%.
- The largest benefit arose from downside protection during unstable market environments, the volatility target of 1.2% daily and regime weight allocation bounds.

The results suggest that complexity metrics contain useful information regarding changes in market structure and can be useful for constructing a resilient portfolio.

6. Conclusion

This project demonstrates that nonlinear complexity measures can be integrated into a practical market-regime detection framework for portfolio management. By combining the Hurst Exponent, Largest Lyapunov Exponent, Sample Entropy, and Autocorrelation, the model identifies persistent, chaotic, and equilibrium market environments and adjusts portfolio allocations accordingly.

The out-of-sample results indicate that regime-aware allocation can improve risk-adjusted performance while substantially reducing drawdowns relative to a passive investment approach. The framework provides a promising foundation for future research involving complex dynamics and more advanced portfolio construction techniques.